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Slot Doubler Pile Installation A Novel Approach Without LBL Positioning

C. F. Amaechi, Brownfield Project & Construction, Shell Nigeria Exploration & Production Company, Eko Atlantic, Lagos, Nigeria; P. Halim, Geomatics, Shell Nigeria Exploration & Production Company, Eko Atlantic, Lagos, Nigeria

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Abstract

This paper discusses a bold and novel approach taken for the installation of the suction pile for the Subsea Slot Doubler in Shell's deepwater Bonga field, offshore Nigeria, using a Remote Operated Vehicle (ROV)-mounted Inertial Navigation System (INS) in the absence of the traditional and well-established Long Baseline (LBL) positioning system. The method utilized INS-supported waypoint referencing to position marker buoys for grid positioning, real-time ROV guidance, and dynamic drift correction to ensure placement accuracy. Field implementation resulted in an installed horizontal deviation of 0.36 meter which was less than the design tolerance of 1.00 meter, approximately 25%-time savings compared to LBL-enabled operations, and reduced survey spread footprint. This paper provides the methodology, operational results, and discusses the implications for lean deep offshore operations not only restricted to learner markets that is defined by simplicity, cost efficiency and repeatability in the region offering the benefit of faster mobilization and easy implementation across various brown fields in the region.

Introduction

The Bonga Field is a subsea deep-water development (average 1000-meter water depth) in the Oilfield Mining License (OML) 118 acreage, that is located 120km south-west of the Niger Delta (See [Fig. 1](#)). And oil production from, and water injection into, the reservoir is facilitated through a network of subsea trees (XTs) and associated hardware, connected to a Floating Production Storage and Offloading facility (FPSO), with first oil having been achieved in 2005.



Figure 1—Picture showing the location of the OML 118 prospect off the coast of Nigeria. Bonga Field Location

With the manifold slots in Bonga having been fully populated, utilize the significant upside potential discovered in the reservoirs, additional wells are being brought on stream using ‘Slot Doublers’ (Fig.2 and 3). These are three-slot cylindrical slot mini manifolds that are tied back to existing manifolds using subsea jumpers, they are compact to fit within an increasingly congested cluster. The installation of the slot doubler subsea is preceded by the installation of a suction pile to a designed location.



Figure 2—Picture (left) showing the suction pile on the installation vessel prior to over boarding by the vessel crane. Slot Doubler Suction Pile.



Figure 3—Picture (right) showing a CAD model view of the slot doubler being installed on the suction pile. Slot Doubler Interface on the Suction Pile.

The accuracy of installation of the suction is critical to avoid well hook up can be achieved with not fatigue failures introduced to the subsea infrastructure. The acceptable tolerance for subsea manifolds in water depths exceeding 200 meters is 1.0 meter for the Horizontal uncertainty and 0.01% Water Depth (WD) for the vertical uncertainty. Further details on the accuracy are provided in subsequent sections.

Given the criticality of the slot doubler and the tight installation tolerances required for this structure and manifolds in general, the infrastructure installation typically relies on LBL (Long Baseline) acoustic positioning which uses a network of transponders for ensuring the correct placement these seabed structures at great water depths (> 200m). And this was the methodology we employed in 2021 when the first slot doubler (being novel at the time) was installed in the Bonga field. Following the success of the first slot doubler installation in 2021 the approach to extend the production plateau by developing nearby opportunities to the production clusters through tie back to existing and fully populated manifolds by the slot doubler is now the base case. With the next installation having been successfully performed in June 2024. However, these LBL systems, while accurate, require considerable mobilization, vendor availability, and offshore vessel time. In this case study, the installation team was faced with a slot doubler installation campaign with no experienced LBL vendor availability in Nigeria in the required window. The critical nature of pile placement and a tight offshore schedule prompted the search for an alternative approach centered on ROV-mounted INS in collaboration with Zupt LLC a US based company delivering Inertial Navigation System base subsea positioning service globally. This paper outlines the methodology, execution, and results achieved using this approach.

Technical Challenge

The subsea campaign required the precise installation of a suction pile within the production manifold – 2 (PM – 2 – Fig.4) cluster location to facilitate bringing on two additional wells – B79 and B43. The suction pile is the foundation for the slot doubler manifold and is first installed on the seabed to refusal.

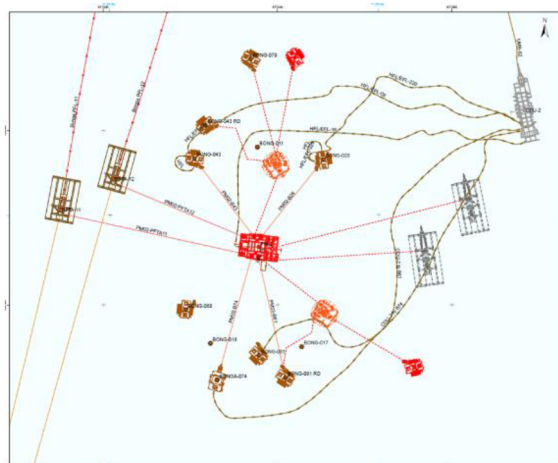


Figure 4—Picture showing the plan layout of the production manifold and well cluster where the suction pile was installed. Production Manifold – 2 Cluster

Due to the unavailability of a proven contractor to execute the LBL survey positioning within the required operational window, installation using LBL was considered as not being a viable option without any major schedule slippage for the project and consequent impact on the annualized production forecast the company has made for the year to underpin the business plans. Thus, the execution team was challenged with developing an alternative installation method that ensured positional accuracy, minimized operational risk, and adhered to the project timeline whilst retaining confidence in the results. Or to defer the installation activity until a competent LBL contractor became available – understandably not a desired option.

Inertial Navigation Systems (INS) had been used in the Bonga field with success for performing well metrologies and well markings in clusters where reference infrastructure existed with known position coordinates. Thus, the INS was proposed as an alternative to the LBL to be used in the slot doubler installation. INS was also considered operationally more efficient and cost competitive as it does not require the same extensive set up as LBL which involves transponder deployment, array set up, acoustic calibration & verification, and extensive vessel time to accomplish the set up tasks. In contrast, INS only required an ROV mounted tool (C-PINS) calibrated against an infrastructure of known location to determine a seabed position.

The proposed INS, however, presented the following technical challenges that had to be overcome.

1. Tracking the position, heading and inclination of the pile during installation. INS do not have this capability as there are no transponders on the suction pile and transponder seabed arrays (see Fig. 5. For a typical array layout) to acoustically track the installation.
2. Accurate determination of the installed position through position monitoring.
3. Lack of suitable analogues to demonstrate success in deep water manifold installation using INS. The INS had not been used in this application (manifold installations) elsewhere that was known to the team or INS contractor, hence the confidence for success remained low requiring the development of a robust installation methodology for this to proceed.

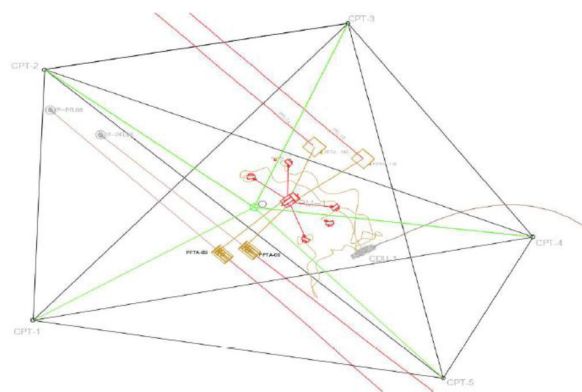


Figure 5—Picture shows a plan layout of the beacon array; lines show the signal communication between them. Typical Seabed Array for LBL Positioning.

The INS Limitations were set out as follows:

What is possible:

- Positioning of marker buoys to $\pm 0.2\text{m}$ relative to known C-PINS alignment
- As-Built position of suction pile to $\pm 0.05\text{m}$ relative to C-PINS alignment sited.
- As-Built attitude (pitch and roll) of suction pile to ± 0.1 degrees with the heading which we provided.

What is not possible:

- Real time positioning, attitude, or yaw of the suction pile throughout the installation
- Provision of As-Built heading of the suction pile. The suction pile design did not incorporate any interface for interfacing with a C-PINS.

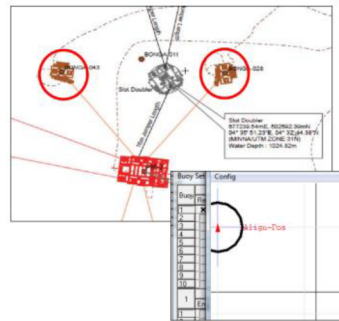
Methodology

The novel and innovative alternative approach to the LBL acoustic positioning installation was developed following a collaborative review between Subsea Intervention and Geomatics teams of Shell Nigeria Exploration and Production Company, Zupt LLC the INS service provider and the Marine Platforms Installation Vessel Offshore Team. The high-level steps involved the utilization of two XLX ROV systems on the installation vessel African Vision as follows.

- Positioning of marker buoys on the seabed using INS to establish a pre-surveyed grid within which the suction pile was to be installed (replicating a method used by drilling rigs for spudding wells during – albeit rig requirements have more generous tolerances).
- Suction pile installation with the ROV docked in, to lock in the orientation
- Close observation of the inclination by the 2nd ROV and adjustments following land out using the rigging configuration and deployment speed, crane positioning.

This was designed to overcome the technical challenges outlined in the previous section. Summarily.

- Alignment location provided through a known location – B-11 well a distance of less than 20m from the buoy location.



- Heading and orientation provided through the ROV. A second ROV provided to increase confidence and maintain monitoring capability.
- Real time tracking – intent assured through use of the marker buoys.

The proposed steps were robustly reviewed and approved by the Shell Geomatics Technical authorities in the regions (Gulf of Guinea and Gulf of America). Following which the project team proceeded to mobilize and execute.

An I – beam was incorporated into the suction pile build to provide a key slot for the ROV to dock in (Fig.6 & 11.). The dimensions and heading were verified as part of the As-Built Dimensional Control Process. The ROV was fitted with a removable ‘fork’ to provide a tight fit to the key slot on the suction pile such that prior to land out on the seabed the ROV could dock in and lock in to orientate the suction pile prior to penetration. The heading was provided by the calibrated Octan Gyro compass on the ROV vehicle.



Figure 6—Picture showing the suction pile installed to refusal on the seabed (ROV footage). Installed suction pile showing the orientation key and ROV platform.

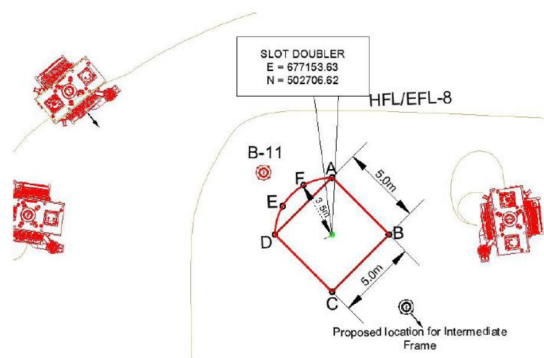


Figure 7—Picture showing the planned position of the marker buoys on the seabed, suction pile to be installed in the center of the square. Buoy Set definition of the survey grid



Figure 8—Picture showing a model view of the C-PINS (cylinder), which is mounted on the buoy jig at the bottom. C-PINS mounted on Buoy Jig Model.



Figure 9—Picture taken by the ROV subsea, in the dark can be seen the marker buoy floating in the deep anchored by the ropes secured to the clump weight (not shown). Marker buoys on the seabed



Figure 10—Picture (left) showing the suction pile on the installation vessel prior to over boarding by the vessel crane. Suction Pile Deployment



Figure 11—Primary ROV system docked on the suction pile during installation. Picture taken by the 2nd ROV system.

The first step of the offshore installation of the suction pile was the accurate installation of 6 marker buoys in a square pattern to mark the proposed installed location of the suction pile (see Fig.7.). The marker buoys provided the pre-surveyed grid to contain the suction pile and served for drift correction. The INS was utilized to install each of the marker buoys.

The square pattern had a dimension of 5×5 m with each buoy a point on the circumference of the semi-circle of a radius of 3.5m (corresponding to the radius of the suction pile) from the proposed center of the slot doubler. The horizontal tolerance for the positioning of the individual maker buoys was defined as a circle of radius 0.15m from target location. Each maker buoy was positioned using INS, to the WGS 84 datum coordinate system (UTM Zone 31 N projection). The coordinates were later converted to MINNA 31N.

The inertial navigation system is a high-precision positioning system that uses motion sensors to determine the position of an ROV mounted C-PINS without requiring constant external signals. This works by obtaining an initial fix subsea on a known structure and then measuring position (Dead Reckoning) via inertial IMU sensors (acceleration is integrated to obtain velocity which is computed to obtain changes in position) as the ROV navigates from point to point (Zupting). This produces to a high degree of accuracy the positional coordinates even in environments where acoustic performance is degraded by 'noise'.

The marker buoy installation was performed by the Zupt Survey team utilizing the following tooling:

1. C-PINS (Fig.8)
2. Buoy Jig (Fig.8)
3. And Rotating Stabs
4. AquaNav Computer

The ROV executed precise maneuvers to calibrate drift and establish waypoints corresponding to the marker buoy location to place each of the 6 marker buoys. Target accuracy for the marker buoy -

- Relative accuracy between marker buoys < 0.05 m
- Relative accuracy to B-11 < 0.2 m

The first step was to establish C-PINS alignment against the well B-11 well head of which the position was known. This was carried out by the ROV placing the C-PINS on the well head receptacle and obtaining heading for 5 rotated positions. C-PINS position was confirmed on the AquaNav topsides and aligned with the known position of B-11. With this task completed the ROV then placed the first buoy in the buoy jig and then moving it to the buoy A location. C-PINS was used to adjust the buoy location and obtain position

loops, and the Buoy Northing and Easting position was computed. This was repeated for Buoys B, C, D, E and F – with all buoys placed relative to the B-11 (alignment position) (Fig.9).

Following completion of the marker buoy positioning using INS and the As-Built survey by the ROV the suction pile was then deployed to locate it within the grid defined by the marker buoys (Fig.12).



Figure 12—Picture taken by the ROV subsea and showing the suction Pile captured within the marker buoys on its way to self penetration.

The suction pile (Fig.10.) was deployed from the deck for the African Vision and installed within the marker buoy grid. The vessel bridge team used live position data from the crane beacons and ROV overlay on the APOS to guide the installation of the suction pile to the target coordinates. Real-time visual confirmation of orientation and verticality was conducted using the visual observation of the 2nd ROV which was free to move between observation of the bull's eye on the pile and a standoff position to observe the entire installation process, which also involved ensuring the suction pile was lowered within the boundaries of the marker buoys. Following touch down on the mud line, load adjustments were made to ensure level, all through the primary ROV remained locked in to ensure accurate heading positioning for the suction pile (Fig.11).

When the installation was completed, INS was used to obtain as – built position data for the suction pile.

Results and Discussion

The operation was completed successfully with a final installation horizontal deviation of 0.36 meters (less than the allowed deviation of 1.0 meters), within acceptable tolerances. ROV observation confirmed pile verticality and embedment. Post-installation INS survey corroborated positional accuracy.

The results of the marker buoy positioning are given in the tables 1 – 2 and Figure 13, which formed the basis for a successful installation of the suction pile, and show that they were within the expected target tolerances (tables 3 – 4, Figure 14)

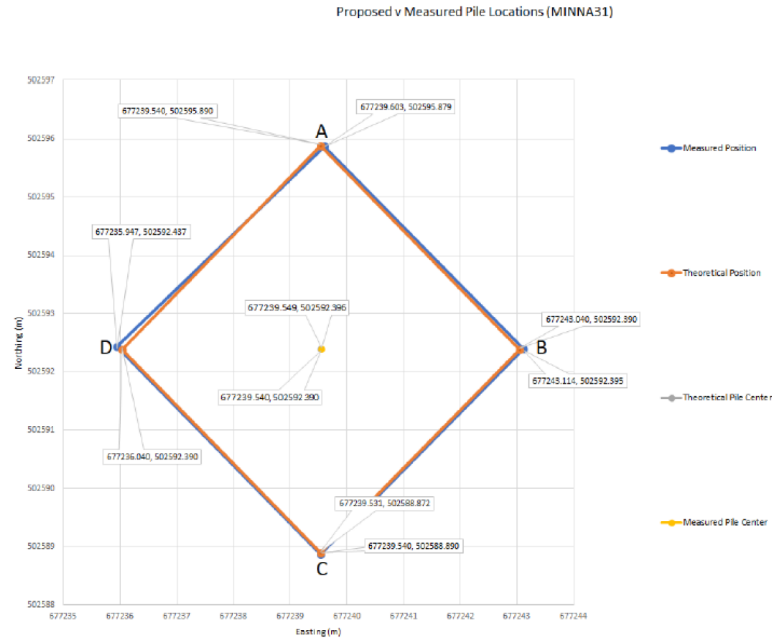
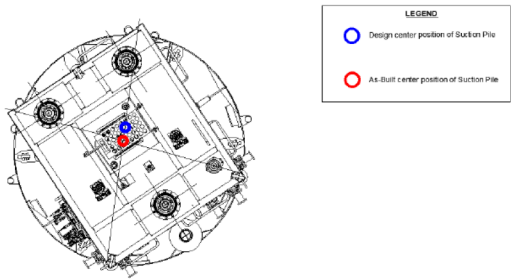


Figure 13—Picture showing the As Built marker buoy lay out (Planned overlaid by actual). Difference in proposed and as-built suction pile buoy locations – Minna Zone 31N

Parameter	Variable Name	Design		As-Built		Zupt Difference	SNEPCo Tolerance	Notes
Horizontal Position	E, N	E	N	E	N	0.36m	+1.00m	Raw receptacle position at center of suction pile relative to B11.

B-11



SUCTION PILE

Figure 14—As-Built Suction Pile Installation Data – Table of As Built Coordinates and plan view of slot doubler on suction pile.

Table 1—As-Built Suction Pile Buoy Locations – MINNA Zone 31N

Suction Pile	Proposed Northing (m)	Proposed Easting (m)	Measured Northing (m)	Measured Easting (m)	Error in Northing (m)	Error in Easting (m)	Linear Error (m)
Buoy A (N)	502595.890	677239.540	502595.879	677239.603	-0.011	0.063	0.064
Buoy B (E)	502592.390	677243.040	502592.395	677243.114	0.005	0.074	0.074
Buoy C (S)	502588.890	677239.540	502588.872	677239.531	-0.018	-0.009	0.020
Buoy D (W)	502592.390	677236.040	502592.437	677235.947	0.047	-0.093	0.104
Centroid	502592.390	677239.540	502592.396	677239.549	0.006	0.009	0.010

Table 2—As-Built Boundary Buoy Location – MINNA Zone 31N

Boundary	Proposed Northing (m)	Proposed Easting (m)	Measured Northing (m)	Measured Easting (m)	Error in Northing (m)	Error in Easting (m)	Linear Error (m)
E (WNW)	502594.140	677236.510	502594.176	677236.536	0.036	0.026	0.044
F (NNW)	502595.420	677237.790	502595.354	677237.753	-0.066	-0.037	0.076

Table 3—As-Built Slot Doubler Installation Data

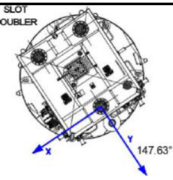
Parameter	Variable Name	Design	As Built	Zupt Difference	Notes	Sketch
Line Azimuth	Heading	147°	147.63°	+3.5°	Line azimuth from center of suction pile to guidepost through fore hub of the slot doubler.	
Slot Doubler Pitch	Inclination (Y)	0°	-2.75°	+2°	Inclination of slot doubler along the Y-axis looking in the direction of the guidepost azimuth: 147.63°	
Slot Doubler Roll	Inclination (X)	0°	+0.94°	-2°	Inclination of slot doubler along the X-axis looking in the direction of the guidepost azimuth	

Table 4—Horizontal deviation tolerance

Description	Absolute Required Tolerance	Actual
Horizontal Uncertainty	1.0m or less	0.36m

The buoy setting timelines were as follows:

- Alignment – 1 hour.
- Buoy placement – 2 hours

For emphasis – no array calibration was needed using this method.

The subsequent installation of the slot doubler on the suction pile further confirmed the accuracy of the suction pile installation.

The results obtained were within the acceptable tolerances defined for our deepwater manifold installations

Tables 3 - 4 show that the installation of the slot doubler as a result was within expected tolerance. Performance of the ROV and being able to utilize two ROV systems was critical to the successful outcome as was the accuracy of the operation of the heave compensated crane from the vessel to accurately land in the target zone as well as using weight offsets to maintain the level of the suction pile. Survey data was reliable, repeatable, and confirmed by the As-Built surveys and a detailed review of the OPLOG data from C-PINS – Zupt LLC also has a stringent quality assurance methodology to ensure data accuracy and quality which comprises of obtaining multiple data points quickly and processing in a mathematical model only data which are deemed reliable.

With regards to schedule, compared to previous campaigns with LBL systems, the INS-led method was 25% faster in execution time.

Table 5—Comparison of operation timeline

Description	2021 Using LBL	2024 using INS
Quayside Calibration	60 hours	0 hours
Vessel transit time to port for quayside calibration	72 hours (executed simultaneously with suction pile mob)	0 hours (not required)
Offshore Execution	184:32 hrs.	111:56 hours

System Performance Insights

The integration and simultaneous operation of the two ROVs was a critical success factor – Primary ROV docked to the suction pile for heading control and positional lock-in. Secondary ROV provided dynamic situation awareness, verifying buoy clearance, crane hook path and pile verticality during touch down.

The two-vehicle approach also enable immediate visual confirmation of pile position relative the marker grid, supporting fine adjustment during the lowering sequence.

Industry Implications

The success of this campaign demonstrates that INS can work in installation of subsea structures as an alternative to LBL methodology, where robust procedural methods are developed and controls implemented. This method therefore can achieve LBL- level accuracy in manifold foundation installations. Which then opens opportunities for leaner, faster mobilizations particularly whereas was the case in Bonga the hardware are mobilized offshore well ahead of the installation window which thus makes a dedicated port call for the mobilization of LBL survey spread very undesirable and the alternative long-term rental of the equipment for a one-off installation campaign also not attractive. And where schedule compression is critical as the INS team can be mobilized as was the case directly offshore.

Furthermore, the absence of acoustic transponder deployment reduced risk and exposure hours, resulting in improved HSE performance. This approach proved effective in constrained environments limited by the constraint on being able to mobilize competent LBL contractors quickly in the region and enabled flexible, rapid deployment.

Conclusions

This project has proven that INS-based subsea installation of suction piles for manifolds presents a reliable alternative to traditional LBL positioning under constrained logistics or schedule. Being technically robust and operational efficient.

Key takeaways / Justifications:

1. Precision achieved: Achieved 0.36m horizontal deviation and < 0.1 -degree inclination error. Exceeding the required deepwater manifold tolerance requirements. These outcomes confirm that INS achieved (and in some respects exceeded) the placement accuracy required and hitherto thought to be only achievable using LBL based campaigns. The independent validation layers (INS As-Built coordinates obtained, ROV As-Built survey) reinforced the confidence in the results and consequently reliability that this method is repeatable. The two production wells in the cluster that were drilled at those locations have been commissioned successfully with the jumper fabrication and installation to connect the wells to the slot doubler being executed without any difficulty or variations from the jumper design tolerances – a fundamental demonstration of success.
2. Efficiency Gains: Reduced execution time by approximately 25% when compared to LBL installation method. Eliminating over 100 hours of quayside calibration and transit activities which also translated to cost savings of approximately six hundred thousand US Dollars on vessel time. This is based on a direct comparison to the 2021 suction pile installation using LBL, where 132 hours in total was required for the calibration, quayside set up and transit activities. This efficiency gains directly translated into a cost saving of approximately six hundred thousand US Dollars due to reduced vessel time, reduced vessel time also provided the benefit of lower greenhouse gas emissions. These are tangible benefits to operators who desire to expand the well count capability at already populated production clusters.

3. Risk Reduction: Lower subsea exposure hours and simplified spread requirements reduces operational risk and HSE outcomes. Justification of this conclusion is drawn from the reduced vessel time (which translates to man-hours), lower foot print of the INS spread which means no lifting operations besides the ROV deployments – less lifting implies fewer chances of lifting and hoisting related incidents such as dropped objects and line of fire accidents.
4. Scalability: Applicable to brownfield environments with known infrastructure references, especially in less mature markets where LBL vendor capacity is constrained.

In conclusion, this Shell Bonga case study provides confidence that INS, when deployed with rigor and supported by well-defined operational controls, is a viable, repeatable, and industry-scalable alternative for precision installation of subsea foundations. Presenting a reliable alternative to traditional LBL positioning and not just under constrained logistics or schedule demands.

Strategic Value for the Industry

The method demonstrated by the Bonga BP & C team and their contractor partners aligns with a push across the industry for lean and safe execution embodied by simplification, performance & standardization, as well as cost avoidance without compromising on Safety and Technical assurance. It offers a blueprint for contingency planning in vessel – based installation campaigns, particularly in emerging or logistically constrained markets.

The conclusion is justified by the results of this campaign, which has successfully expanded the use case and built on the successes of the well jumper metrology campaign using INS in the Bonga field for the past five years.

Outlook

The team is looking at applying the methodology to installation of larger frames and tie-in spools in a brownfield setting. Operators should also consider incorporating INS methodology into contingency planning, with future advancements focusing on resident vehicle integration with very tight installation tolerances to enable faster deployment to execution times.

Acknowledgments

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Nomenclature

INS	– Inertial Navigation System
LBL	– Long Baseline Acoustic Positioning
ROV	– Remote Operated Vehicle
SNEPCo	– Shell Nigeria Exploration and Production Company

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